Quality Information Document

Ocean Monitoring Indicator (OMI) of the Copernicus Marine Environment Monitoring Service

Evolution over 1993-2023 of the sub-surface temperature (0-250m) anomaly averaged in the NINO3.4 box

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CHANGE RECORD

When the quality of the products changes, the QUID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

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I INTRODUCTION

I.1 Product name

GLOBAL_OMI_CLIMATE-VARIABILITY_NINO34_Tzt_anom

I.2 Indicator evaluation method

The average inside the NINO 3.4 box (170°W-120°W, 5°S-5°N) of subsurface temperature over the period 1993-2023 minus the 1993-2014-time average is evaluated to obtain a profile time series "depth-time" plot of subsurface temperature anomaly, expressed in °C.

A multi-product approach is used, meaning that the subsurface anomaly is computed for 43 different global 3D temperature estimates and for their ensemble mean (GREP). The figure only displays the ensemble mean results but the file contains all the members.

I.3 CMEMS products used as input for the computations

Product name (and Type)	Documentation
GLOBAL_MULTIYEAR_PHY_ENS_001_031	QUID: CMEMS-GLO-QUID-001-031.pdf PUM: CMEMS-GLO-PUM-001-031.pdf (copernicus.eu)
INSITU_GLO_TS_REP_OBSERVATIONS_013_001_b	QUID: Foreword (copernicus.eu) PUM: CMEMS-INS-PUM-013-001-b.pdf (copernicus.eu)

The 3 reanalyses of GREP (GLORYS, CGLORS, ORAS5) GLOBAL_MULTIYEAR_PHY_ENS_001_031 use the NEMO ocean model with assimilation of in situ and satellite observations.

They are evaluated against INS TAC in situ observations INSITU_GLO_TS_REP_OBSERVATIONS_013_001_B in the NINO3 and NINO4 areas.

II SCIENTIFIC VALIDATION

More details and full scientific evaluation can be found in the CMEMS Ocean State Report #2 (Gasparin et al 2018). The nino3.4 subsurface temperature OMI is based on the "multi-product" approach as introduced in the second issue of the Ocean State Report (CMEMS OSR#2, 2018).

The scientific validation of this OMI relies on the study of the consistency in between the 4 different estimates (standard deviation), and the analysis of the average departures of the reanalysis GLORYS2V4 from in situ temperature observations in the NINO3 and NINO4 boxes.

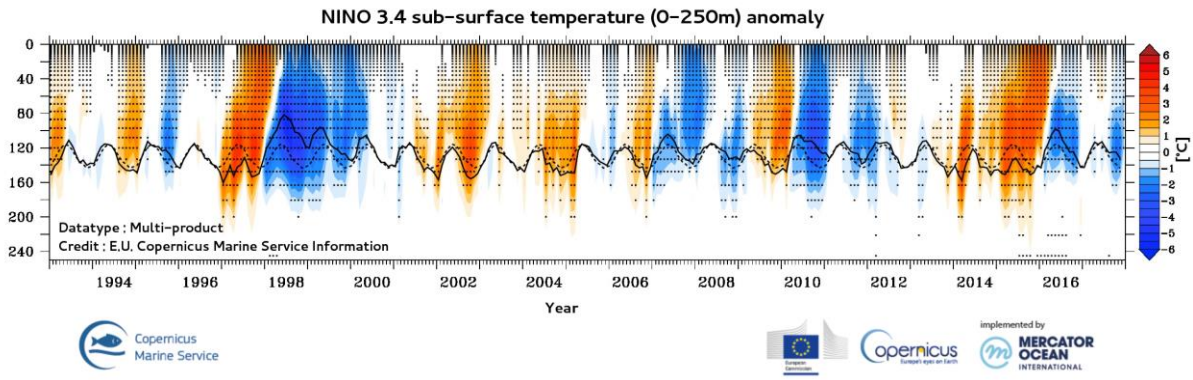

Figure 1: Temperature anomalies, from the 1993-2014 annual mean, area-averaged in the Niño box 3.4 from the multi-product approach. The thick black line indicates the position of the isopycnal 1025 kg/m^3, and the dashed black line indicates its climatological position. Black dots mark the areas where the anomaly is at least twice the standard deviation of temperature anomalies from the four multi-product members (e.g. the standard deviation is less than 0.6°C). Units are °C.

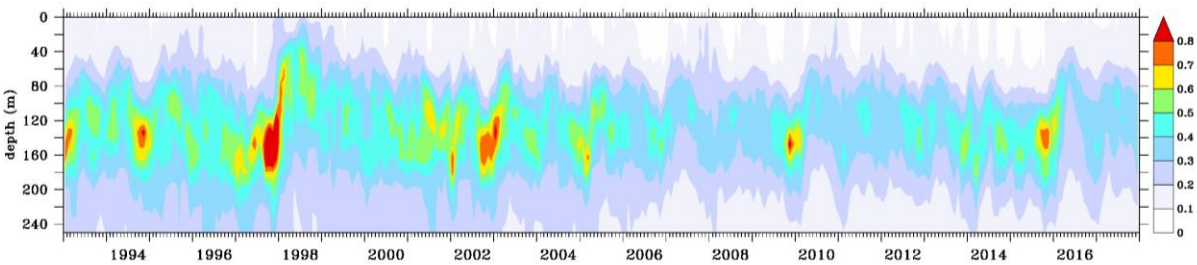

Figure 2 : Standard deviation of temperature (°C) in between the four memebers of GREP (GLORYS, CGLORS, ORAS5, FOALM) averaged in the Niño box 3.4

The GREP multi-product ensemble mean and its associated standard deviation as available in the CMEMS catalogue were used to compute the temperature anomalies (Figure 1 & 2). As

can be seen in Figure 1, the anomalies are of the order of 1 to 6 °C and are almost all the time significant, as they are at least twice larger than the standard deviation of temperature in between individual members, which is shown in Figure 2, and which seldom exceeds 1°C, mostly in the thermocline area. The strong ENSO signal of 1997/1998 clearly stands out, while other remarkable El Nino events (2015, 2002 and 2009) and La Nina events (2007, 2010, 2016) are also detected.

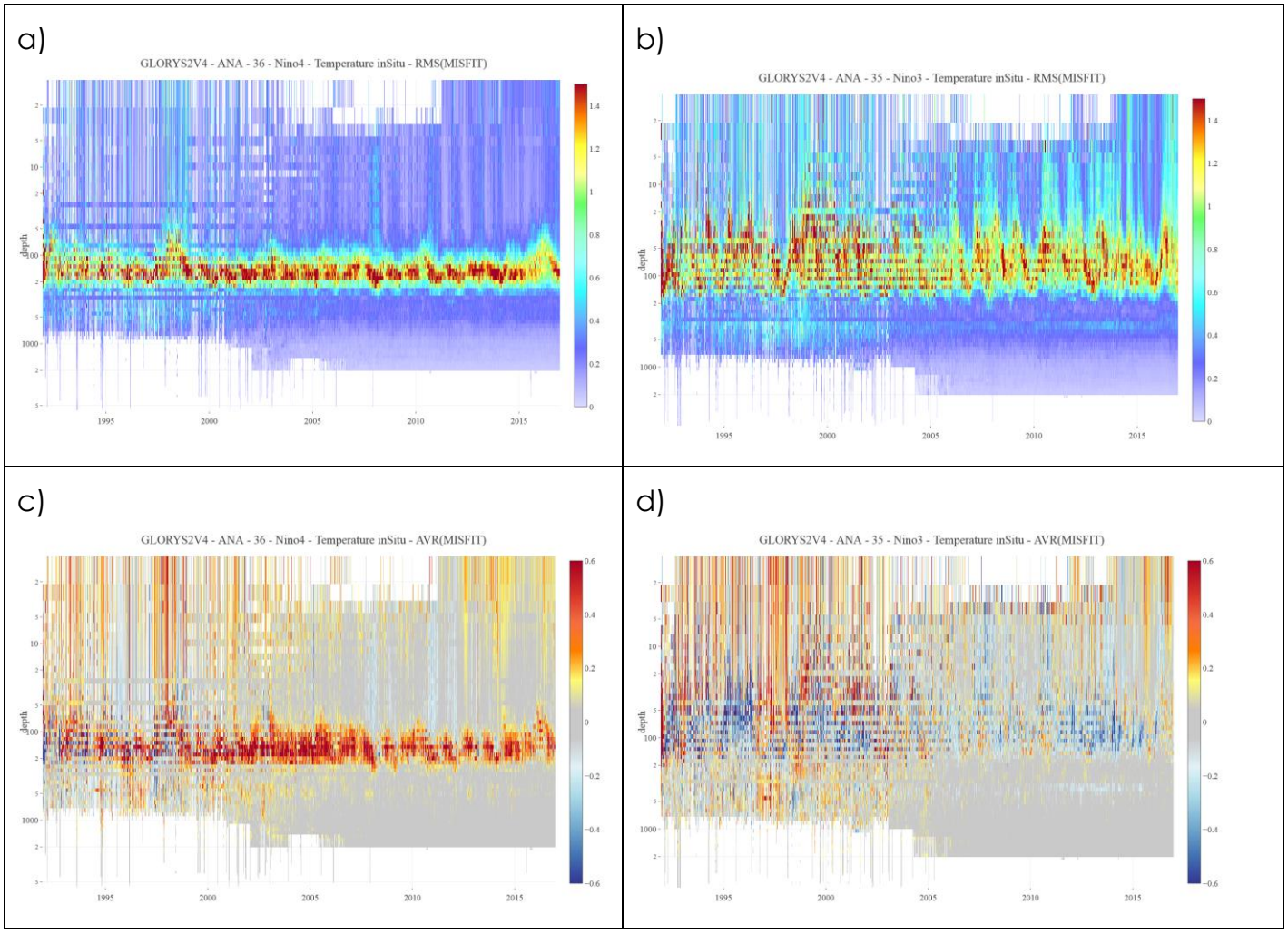

Figure 3 : uncertainty of global reanalyses in the NINO4 (left column a and c) and NINO3 (right column b and d) areas over the 1993-2016 period, illustrated using vertical profiles of average (lower panel c and d) and root mean square (upper panel a and b) deviations of GLORYS with respect to reference in situ temperature observations

All four members of GREP (GLORYS, CGLORS, ORAS5, FOAM) assimilate in situ temperature and salinity profiles among other observations, and are close to these observations on average. They reproduce well the tropical oceans variability and are used as oceanic initial conditions for seasonal prediction models (coupled ocean-atmosphere models). As illustrated with GLORYS results in Figure 3, the reanalysis departures from in situ temperature profiles is

strong in the thermocline (of the order of 1-1.5°C), where internal wave processes can generate large high frequency fluctuations that are not reproduced by the models. A cold bias is diagnosed in the thermocline in GLORYS in the central Pacific (nino 4 area) which suggests that the thermocline is too shallow in that area, but which does not prevent from reproducing correctly the strong variability of temperature in the 0-250m layer. Note that the RMS departures in Figure 3 are larger than the spread in Figure 2 because the RMS of Figure 3 is computed in the observations' space, while the spread (standard deviation) of Figure 2 is computed between monthly means of the four reanalyses.

III REFERENCES

Gasparin et al., 2018: The 2015 El Nino event. In: The Copernicus Marine Environment Monitoring Service Ocean State Report, issue 2, Journal of Operational Oceanography;